SYEDA UMAIMA TAMKEEN

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ACADEMIC PROFILE

Artificial Intelligence graduate, university lecturer, and published researcher with experience in machine learning, deep learning, natural language processing, and applied AI. Research interests include responsible AI, algorithmic fairness in healthcare, and practical AI systems. Experienced in designing and delivering university-level technical instruction, mentoring students, and developing machine learning applications.

EDUCATION

Bachelor of Science in Artificial Intelligence | Dawood University of Engineering & Technology, Karachi | November 2021 – June 2025

CGPA: 3.4

RESEARCH INTERESTS

Responsible and Fair AI | Machine Learning and Deep Learning | Healthcare AI | Natural Language Processing | Computer Vision

PEER-REVIEWED PUBLICATION

Mitigating Bias in Healthcare Algorithms Through Fairness Constraints: A Multi-Site Validation Study with Real-World Clinical Data

International Journal of Research and Analytical Reviews (IJRAR), Vol. 12, Issue 3, September 2025. ISSN: 2348-1269. Reference ID: IJRAR25D1245.

- Studied algorithmic fairness in healthcare AI using more than 156,000 patient records from multiple sites.
- Implemented adaptive fairness constraints; reported a 42% reduction in demographic bias while maintaining predictive performance (AUC = 0.851).
- Worked with healthcare professionals on clinical validation; reported a 73% improvement in fairness-rated predictions.
- Proposed a continuous monitoring framework for fairness in clinical machine learning systems.

TEACHING EXPERIENCE

AI Instructor | Bano Qabil | August 2026 – Present

- Teach foundational AI concepts and practical applications through structured, hands-on instruction.
- Mentor learners on AI projects and development of applied technical skills.

Visiting Faculty Lecturer — Data Structures & Algorithms | Newports Institute of Communications and Economics, Karachi | February 2026 – Present

- Teach undergraduate data structures, algorithms, and programming implementation.
- Develop lectures, lab materials, and assessments; support students in algorithm design and academic projects.

Teaching Associate — Artificial Intelligence & Python Programming | Aptech Learning, Karachi | August 2025 – December 2025

- Delivered lectures and practical sessions in AI, machine learning, and Python programming.
- Designed lab activities, supervised projects, and mentored students on technical and academic development.

RESEARCH AND TECHNICAL EXPERIENCE

Artificial Intelligence Engineer (Remote) | Krank, London, United Kingdom | June 2025 – November 2025

- Developed ML approaches supporting equipment inspection and predictive maintenance workflows.
- Integrated TensorFlow and PyTorch modules for anomaly detection and equipment valuation prediction.
- Developed data pipelines and collaborated with remote cross-functional teams on analytics delivery.

Artificial Intelligence Intern | Kangaroo Ventures, USA | 2025

- Built applied AI projects including a recommender system, NLP resume parser, traffic-sign classifier, and Transformer-based chatbot.
- Performed sentiment analysis on social media text.

Additional Machine Learning and Data Analysis Internships | 2023–2024

Growth Intern (India, 2024): fraud detection and age/gender classification; **Digital Empowerment Network** (Pakistan, 2024): regression, classification, churn, and image recognition; **TechnoHacks EduTech** (India, 2023): sales prediction and feature engineering.

SELECTED ACADEMIC PROJECTS

- Gesture Vocalizer Assistive Communication Device — Final Year Project.
- Handwritten Optical Character Recognition (OCR).
- Diabetes Risk Prediction and Multi-Disease Diagnostic Tool using machine learning.

ACADEMIC LEADERSHIP AND OUTREACH

Founder & President, AI Society | Dawood University | 2023–2024

- Organized and led AI workshops, boot camps, and hackathons involving more than 300 participants from regional universities.

ADDITIONAL PROFESSIONAL EXPERIENCE

Project Manager | Fumograph, Karachi | December 2025 – June 2026

Coordinated client requirements, project planning, cross-functional teams, and delivery of web, design, and multimedia solutions; supported business development and client relationships.

TECHNICAL COMPETENCIES

Programming and Analytics: Python, data structures and algorithms, data analysis, business intelligence.

Artificial Intelligence: machine learning, deep learning, NLP, Transformers, computer vision, predictive modeling.

Frameworks: TensorFlow, PyTorch. **Academic Practice:** curriculum development, lab design, assessment, mentoring.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

Google: Advanced Data Analytics; AI Essentials; Business Intelligence; Data Analytics; Project Management; Python; IT Automation.

Certified Data Scientist — PITP (NED Academy).